

6(a)	particle with no internal structure / particle which cannot be broken down into anything smaller	A1
6(b)	charges: $u = (+)\frac{2}{3}(e)$ or $d = -\frac{1}{3}(e)$ or $s = -\frac{1}{3}(e)$	C1
	$(+)\frac{2}{3}(e) - \frac{1}{3}(e) - \frac{1}{3}(e) = 0(e)$	A1
6(c)(i)	• same/equal mass • same/equal (magnitude of) charge • both fundamental (particles) • opposite (sign of) charge • one is matter and the other is antimatter <i>Any two points, 1 mark each.</i>	B2
6(c)(ii)	neutron/baryon consists of three quarks	B1
	pion/meson consists of one quark and one antiquark	B1